

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity. (BL2, BL3)*

Activity Tool : MCQ Test (Low)
Assessment Tool Weightage: 30%

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Which of the following is true about 1NF? (a) Allows multi-valued attributes (b) Eliminates partial dependencies (c) All attributes must be atomic (d) Eliminates transitive dependencies	BL2 – Understand	5
2	A relation is in 2NF if it is in 1NF and: (a) Has no transitive dependencies (b) Has no partial dependencies on primary key (c) Every attribute is a prime attribute (d) Has no multi-valued dependencies	BL2 – Understand	5
3	Which normal form deals with multi-valued dependencies? (a) 2NF (b) 3NF (c) BCNF (d) 4NF	BL2 – Understand	5
4	BCNF is a stricter version of: (a) 1NF (b) 2NF (c) 3NF (d) 4NF	BL2 – Understand	5
5	Functional dependency $X \rightarrow Y$ means: (a) Y determines X (b) X determines Y (c) X and Y are unrelated (d) Both determine each other	BL2 – Understand	5
6	Which of the following anomaly is NOT associated with unnormalized relations? (a) Insertion anomaly (b) Deletion anomaly (c) Update anomaly (d) Retrieval anomaly	BL2 – Understand	5
7	A superkey is: (a) A minimal set of attributes that uniquely identifies tuples (b) Any set of attributes that uniquely identifies tuples (c) An	BL2 – Understand	5

	attribute that depends on a non-prime attribute (d) A foreign key reference		
8	If R(A,B,C) has FD: $A \rightarrow B$ and $B \rightarrow C$, then C is: (a) Partially dependent on A (b) Fully dependent on A (c) Transitively dependent on A (d) Independently determined	BL2 – Understand	5
9	5NF is also known as: (a) Domain-Key Normal Form (b) Boyce-Codd Normal Form (c) Project-Join Normal Form (d) Element Normal Form	BL2 – Understand	5
10	Lossless-join decomposition ensures: (a) No data is lost during decomposition (b) All FDs are preserved (c) The relation is in BCNF (d) No null values exist	BL2 – Understand	5

Code of Conduct:

I ___S. Jaya Shree(192524146)___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S. Jaya Shree

RUBRICS FOR CALCULATION:

MCQ					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

Q1. Which of the following is true about 1NF?

Options:

- (a) Allows multi-valued attributes
- (b) Eliminates partial dependencies
- (c) All attributes must be atomic
- (d) Eliminates transitive dependencies

Answer: (c) All attributes must be atomic

Explanation:

First Normal Form (1NF) is the first stage of database normalization. A relation is in 1NF when:

1. Each cell contains only one value.
2. All attributes contain atomic values.
3. There are no repeating groups or multi-valued attributes.

Example:

Not in 1NF:

Student ID	Name	Phone
101	Jaya	98765, 87654

The Phone attribute contains multiple values.

In 1NF:

Student ID	Name	Phone
101	Jaya	98765
101	Jaya	87654

Now each cell contains a single atomic value.

Conclusion:

1NF ensures that all attributes contain atomic (single) values.

Q2. A relation is in 2NF if it is in 1NF and:

Options:

- (a) Has no transitive dependencies
- (b) Has no partial dependencies on primary key
- (c) Every attribute is a prime attribute
- (d) Has no multi-valued dependencies

Answer: (b) Has no partial dependencies on primary key

Explanation:

Second Normal Form (2NF) is achieved when:

1. The relation is already in 1NF.
2. There is no partial dependency of any non-key attribute on a part of the primary key.

A partial dependency occurs when a non-key attribute depends on only one part of a composite primary key, instead of the complete primary key.

Example:

Consider:

Student_ID	Course_ID	Student_Name	Course_Name
101	C01	Jaya	DBMS
102	C02	Ravi	Python

Primary Key = (Student_ID, Course_ID)

- Student_ID $\rightarrow$ Student_Name
- Course_ID $\rightarrow$ Course_Name

Here, Student_Name depends only on Student_ID and Course_Name depends only on Course_ID. Therefore, partial dependencies exist.

To achieve 2NF, the table should be decomposed into separate tables.

Conclusion:

2NF removes partial dependencies and ensures that non-key attributes depend on the entire primary key.

Q3. Which normal form deals with multi-valued dependencies?**Options:**

- (a) 2NF
- (b) 3NF
- (c) BCNF
- (d) 4NF

Answer: (d) 4NF

Explanation:

Fourth Normal Form (4NF) deals with multi-valued dependencies (MVDs). A multi-valued dependency occurs when one attribute determines multiple independent values of another attribute.

Example:

A student may have multiple hobbies and multiple skills. If both are stored in the same table, unnecessary combinations may occur.

4NF separates these independent multi-valued facts into different relations.

Conclusion:

4NF = No unwanted multi-valued dependencies

Q4. BCNF is a stricter version of:**Options:**

- (a) 1NF
- (b) 2NF
- (c) 3NF
- (d) 4NF

Answer: (c) 3NF

Explanation:

BCNF (Boyce-Codd Normal Form) is a stronger version of 3NF. In BCNF, for every functional dependency $X \rightarrow Y$, X must be a superkey.

BCNF removes certain types of redundancy that may still exist in a relation satisfying 3NF.

Example:

If $A \rightarrow B$, then A must be a superkey for the relation to satisfy BCNF.

Conclusion:

BCNF = Stricter version of 3NF

Q5. Functional dependency $X \rightarrow Y$ means:**Options:**

- (a) Y determines X
- (b) X determines Y
- (c) X and Y are unrelated
- (d) Both determine each other

Answer: (b) X determines Y

Explanation:

A functional dependency $X \rightarrow Y$ means that the value of X uniquely determines the value of Y. If two rows have the same value of X, they must have the same value of Y.

Example:

Student_ID $\rightarrow$ Student_Name

A particular Student_ID identifies one Student_Name.

Conclusion:

$X \rightarrow Y$ means X determines Y.

Q6. Which of the following anomaly is NOT associated with unnormalized relations?**Options:**

- (a) Insertion anomaly
- (b) Deletion anomaly
- (c) Update anomaly
- (d) Retrieval anomaly

Answer: (d) Retrieval anomaly

Explanation:

Poorly designed or unnormalized relations can cause three major anomalies:

1. Insertion anomaly – difficulty inserting new information.
2. Deletion anomaly – deleting information may accidentally remove other useful information.
3. Update anomaly – the same information may need to be updated in several places.

Retrieval anomaly is not one of the standard three modification anomalies.

Conclusion:

Answer = Retrieval anomaly

Q7. A superkey is:

Options:

- (a) A minimal set of attributes that uniquely identifies tuples
- (b) Any set of attributes that uniquely identifies tuples
- (c) An attribute that depends on a non-prime attribute
- (d) A foreign key reference

Answer: (b) Any set of attributes that uniquely identifies tuples

Explanation:

A superkey is a set of one or more attributes that can uniquely identify every tuple (row) in a relation. It may contain extra attributes.

Example:

In a Student table:

Student_ID $\rightarrow$ Superkey

Student_ID + Student_Name $\rightarrow$ Also a Superkey

A **candidate key** is a minimal superkey.

Conclusion:

Superkey = Any attribute set that uniquely identifies a row

Q8. If R(A,B,C) has FD: $A \rightarrow B$ and $B \rightarrow C$, then C is:

Options:

- (a) Partially dependent on A
- (b) Fully dependent on A
- (c) Transitively dependent on A
- (d) Independently determined

Answer: (c) Transitively dependent on A

Explanation:

Given:

$A \rightarrow B$

$B \rightarrow C$

Therefore:

$A \rightarrow B \rightarrow C$

C depends on A through B. This is called a transitive dependency.

Example:

Student_ID $\rightarrow$ Department_ID

Department_ID $\rightarrow$ Department_Name

Therefore:

Student_ID $\rightarrow$ Department_Name

Department_Name is transitively dependent on Student_ID.

Conclusion:

$A \rightarrow B$ and $B \rightarrow C = C$ is transitively dependent on A

Q9. 5NF is also known as:

Options:

- (a) Domain-Key Normal Form
- (b) Boyce-Codd Normal Form
- (c) Project-Join Normal Form
- (d) Element Normal Form

Answer: (c) Project-Join Normal Form

Explanation:

Fifth Normal Form (5NF) is also called Project-Join Normal Form (PJ/NF).

5NF deals mainly with join dependencies. It ensures that a relation cannot be further decomposed into smaller relations without losing information or creating incorrect combinations when the relations are joined.

Example:

A complex relation involving Student, Course, and Teacher may be decomposed into smaller relations and reconstructed using joins.

Conclusion:

5NF = Project-Join Normal Form (PJ/NF)

Q10. Lossless-join decomposition ensures:

Options:

- (a) No data is lost during decomposition
- (b) All FDs are preserved
- (c) The relation is in BCNF
- (d) No null values exist

Answer: (a) No data is lost during decomposition

Explanation:

A lossless-join decomposition means that when a relation is divided into two or more relations, the original relation can be reconstructed exactly by joining them.

It ensures that:

- No original information is lost.
- No extra incorrect tuples are generated.
- The original relation can be recovered.

Example:

Original:

Student(Student_ID, Name, Department)

Decomposed into:

Student(Student_ID, Name)

Department(Student_ID, Department)

Joining these tables using Student_ID should produce the original information.

Conclusion:

Lossless Join = Original data can be recovered without loss.